Composite games evolutionary dynamics in finite populations.

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Abstract

Final year masters project. Replicator dynamics in rock paper scissors (RPS) and composition of RPS and the snowdrift game (SD) in both the infinite and finite population cases. Payoff matrices and replicator derivations, stochastic simulations and drift analysis under different interaction process. Extensive simulation tool for graphing and animation of the different processes and large ensemble averages for general games. Derivation and analytical solution for observation variable H and exact calculation of critical population sizes under different parameters.

This is the extended report containing full derivations of all formulas and more detail about simulation results and software produced.

1 Introduction

1.1 Background and Motivation

Further the work on drift reversal (battle of the sexes, RPS). Double drift reversal? Multiple games interacting. Falling in and out of stable RPS cyclic behaviour.

1.2 Deterministic and Stochastic Evolutionary Dynamics

Infinite population replicator dynamics, stochastic equations for finite populations.

1.3 Cyclic and Composite Games

RPS game, adding our 4th stategy.

This project focuses on a specific composite evolutionary game formed by combining a cyclic Rock-Paper-Scissors (RPS) subgame with a Snowdrift (SD) component.

1.4 Aims of the project

The aim of this project is to analyse evolutionary dynamics in finite populations for a specific composite game combining an RPS subgame with a Snowdrift component, focussing on the effects of stochastic drift, microscopic interaction processes, payoff parameters, and the effect of population size. This will involve the development of an efficient simulation and visualisation tool for analyzing symmetric games in finite populations.

1.5 Objectives

- Formulate the composite RPS-SD game payoff structure and its deterministic replicator dynamics, including fixed points and stability.
- Implement efficient stochastic simulations of composite game (and general symmetric games) for differerent microscopic interaction processes (Moran, Local update, Fermi).
- Define an observation variable, H[4], for the game components, H_{SD}, H_{RPS}, H_4, and compute the expected drift $\langle \Delta H \rangle$ both numerically from simulations and analytically.

- Analyse drift and trajectory behaviour across interaction processes, selection strengths, and payoff parameters.
- Calculate critical population sizes associated with drift reversal[4] under different conditions.
- Develop visualisation tools (time series and simplex-based plots) to compare stochastic and deterministic trajectories and analse drift.

2 Literature Review

2.1 Evolutionary Game Theory

2.2 Replicator Dynamics in Infinite Populations

2.3 Rock-Paper-Scissors and Cyclic Dominance

2.4 Finite Population Effects

2.5 Microscopic Interaction Processes

2.5.1 Moran Process

2.5.2 Local Update Process

2.5.3 Fermi Process

2.6 Drift and Observation Variables

3 Standard RPS

The standard (general) rock paper scissors (RPS) payoff matrix is defined as:

$$\begin{bmatrix} a & c & b \\ b & a & c \\ c & b & a \end{bmatrix} \tag{1}$$

Let x, y, and z represent the fraction of players using rock, paper, and scissors respectively. Since z can be eliminated, we define:

$$z = 1 - x - y$$

Expected payoffs for each strategy can then be defined as follows:

$$\pi_R = ax + b\left(-x - y + 1\right) + cy \tag{2}$$
$$\pi_P = ay + bx + c\left(-x - y + 1\right) \tag{3}$$
$$\pi_S = a\left(-x - y + 1\right) + by + cx \tag{4}$$

The replicator equations can be defined (ref Evo games and pop dynamics hofbauer) for (x, y) using the standard formula:

$$\dot{x}_i = x_i(\pi_i(x) - \langle \pi(x) \rangle) \tag{5}$$

where the average payoff (or mean fitness) in the population is:

$$\langle \pi(x) \rangle = x\pi_R + y\pi_P + z\pi_S$$

This leads to the following replicator equations for x and y:

$$\begin{aligned} \dot{x} = {} & xy\left(ax - ay - bx + b\left(-x - y + 1\right) + cy - c\left(-x - y + 1\right)\right) \\ & + x\left(-x - y + 1\right)\left(ax - a\left(-x - y + 1\right) - by + b\left(-x - y + 1\right) - cx + cy\right) \end{aligned} \tag{6}$$

$$\begin{aligned} \dot{y} = {} & xy\left(-ax + ay + bx - b\left(-x - y + 1\right) - cy + c\left(-x - y + 1\right)\right) \\ & + y\left(-x - y + 1\right)\left(ay - a\left(-x - y + 1\right) + bx - by - cx + c\left(-x - y + 1\right)\right) \end{aligned} \tag{7}$$

Defining $F = \dot{x}, G = \dot{y}$, the following derivatives form the Jacobian used to determine the stability of the system:

$$\begin{aligned}
\frac{dF}{dx} &= xy(a-2b+c) + x(2a-b-c)(-x-y+1) \\
&\quad - x\left(ax - a(-x-y+1) - by + b(-x-y+1) - cx + cy\right) \\
&\quad + y\left(ax - ay - bx + b(-x-y+1) + cy - c(-x-y+1)\right) \\
&\quad + (-x-y+1)\left(ax - a(-x-y+1) - by + b(-x-y+1) - cx + cy\right) \qquad (8) \\
\frac{dF}{dy} &= xy(-a-b+2c) + x(a-2b+c)(-x-y+1) \\
&\quad + x\left(ax - ay - bx + b(-x-y+1) + cy - c(-x-y+1)\right) \\
&\quad - x\left(ax - a(-x-y+1) - by + b(-x-y+1) - cx + cy\right) \qquad (9) \\
\frac{dG}{dx} &= xy(-a+2b-c) + y(a+b-2c)(-x-y+1) \\
&\quad + y(-ax + ay + bx - b(-x-y+1) - cy + c(-x-y+1)) \\
&\quad - y(ay - a(-x-y+1) + bx - by - cx + c(-x-y+1)) \qquad (10) \\
\frac{dG}{dy} &= xy(a+b-2c) + x\left(-ax + ay + bx - b(-x-y+1) - cy + c(-x-y+1)\right) \\
&\quad + y(2a-b-c)(-x-y+1) \\
&\quad - y\left(ay - a(-x-y+1) + bx - by - cx + c(-x-y+1)\right) \\
&\quad + (-x-y+1)\left(ay - a(-x-y+1) + bx - by - cx + c(-x-y+1)\right) \qquad (11)
\end{aligned}$$

The Jacobian is as follows:

$$J = \begin{bmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} \\ \frac{\partial G}{\partial x} & \frac{\partial G}{\partial y} \end{bmatrix}$$

Eigenvalues at the standard RPS game (a = 0, b = 1, c = -1):

$$\left\{ -\frac{1}{2}\sqrt{16x^2 + 16xy - 16x + 16y^2 - 16y + 4}, \quad \frac{1}{2}\sqrt{16x^2 + 16xy - 16x + 16y^2 - 16y + 4} \right\}$$

At the internal fixed point of $(x, y) = (\frac{1}{3}, \frac{1}{3})$, $\lambda_{1,2} = \left\{ -\frac{\sqrt{3}}{3}i, \frac{\sqrt{3}}{3}i \right\}$

No real parts of any eigenvalues, and imaginary components exist therefore neutral stability in standard RPS. These eigenvalues are the solution at the internal fixed point of the standard RPS game. They tell us that the game follows neutral oscillations about the internal fixed point (in the deterministic dynamics) as they contain no real parts (poincare diagram).

Diagram here of the deterministic trajectory for RPS.

4 Augmented RPS

The augmented game can be described by the following payoff matrix.

$$\begin{bmatrix} a & c & b & \gamma \\ b & a & c & \gamma \\ c & b & a & \gamma \\ a+\beta & a+\beta & a+\beta & 0 \end{bmatrix} \qquad (12)$$

$$\begin{bmatrix} \langle RPS \rangle & \gamma \\ \beta & 0 \end{bmatrix}$$

Where the strategies are labelled as R,P,S (as before) and L (the loner strategy). From this the Jacobian matrix can be derived with dimensions 3×3 after using the trick to remove one replicator equation. (q = 1 - x - y - z)

4.1 Payoffs

Firstly the payoffs are defined for the augmented game:

$$\pi_R = ax + bz + cy + \gamma(-x - y - z + 1) \tag{13}$$
$$\pi_P = ay + bx + cz + \gamma(-x - y - z + 1) \tag{14}$$
$$\pi_S = az + by + cx + \gamma(-x - y - z + 1) \tag{15}$$
$$\pi_L = x(a + \beta) + y(a + \beta) + z(a + \beta) \tag{16}$$

4.2 Replicator equations

From these payoffs, the standard payoff difference replicator equations can be defined, note again we need only 3 for the 4 strategy game as the fourth strategy can be elimated using the same trick as RPS, reducing the complexity slightly.

$$\begin{aligned}\dot{x} = {} & x(ax + bz + cy + \gamma(-x - y - z + 1) - x(ax + bz + cy + \gamma(-x - y - z + 1)) \\ & - y(ay + bx + cz + \gamma(-x - y - z + 1)) - z(az + by + cx + \gamma(-x - y - z + 1)) \\ & - (x(a + \beta) + y(a + \beta) + z(a + \beta))(-x - y - z + 1))\end{aligned} \tag{17}$$

$$\begin{aligned}\dot{y} = {} & y\big(ay + bx + cz + \gamma(-x - y - z + 1) - x(ax + bz + cy + \gamma(-x - y - z + 1)) \\ & - y(ay + bx + cz + \gamma(-x - y - z + 1)) - z(az + by + cx + \gamma(-x - y - z + 1)) \\ & - (x(a + \beta) + y(a + \beta) + z(a + \beta))(-x - y - z + 1)\big)\end{aligned} \tag{18}$$

$$\begin{aligned}\dot{z} = {} & z\big(az + by + cx + \gamma(-x - y - z + 1) - x(ax + bz + cy + \gamma(-x - y - z + 1)) \\ & - y(ay + bx + cz + \gamma(-x - y - z + 1)) - z(az + by + cx + \gamma(-x - y - z + 1)) \\ & - (x(a + \beta) + y(a + \beta) + z(a + \beta))(-x - y - z + 1)\big)\end{aligned} \tag{19}$$

Reduced the dimensions to just 3 replicator equations in x, y and z as q = 1 - x - y - z.
Jacobian formed as before, $F = \dot{x}, G = \dot{y}, P = \dot{z}$

$$J = \begin{bmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} & \frac{\partial F}{\partial z} \\ \frac{\partial G}{\partial x} & \frac{\partial G}{\partial y} & \frac{\partial G}{\partial z} \\ \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} & \frac{\partial P}{\partial z} \end{bmatrix}$$

Once again the eigenvalues of this Jacobian matrix can be determined and tell us about the stability of the system of replicator equations derived for the combined game. I have computed these valuse numerically for a particular instance of the game below.

4.3 Stability

Eigenvalues computed using Sympy python library, simplified by hand - floating point precision issue starting to come into play but values are clear (0, 2/9, -2/3 etc).

With a = 0, b = 1, c = -1, $\gamma = 0.2, \beta = 0.1$ we have an internal fixed point at $(2/9, 2/9, 2/9, 1/3)$ (see matrix 52) with eigenvalues (Re, Im):

$$\Big\{(0,\ 0.222222222222222\sqrt{3}), (-0.0666666666666667,\ 0), (0,\ -0.222222222222222\sqrt{3})\Big\} \tag{20}$$

All the real parts are <= 0, indicating neutral stability (oscillations at fixed distance around the interior fixed point) also visible in the simulations 1.

See figure 1 for the 4 simplex plot of these dynamics for the local update and Moran processes aswell as numerical solution of the replicator system of ODE's derived from the standard replicator equation (local update in the limit $N \to \infty$). With a suitably large population the local and Moran processes are following the deterministic trajectory but with added noise.

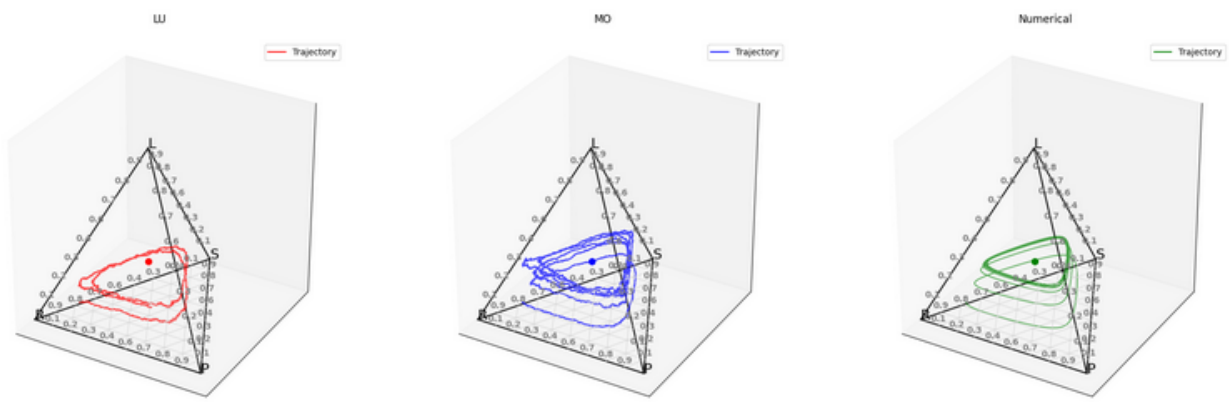


Figure 1: Simulated trajectories for this instance of the game under Local update and Moran process and the deterministic solution from the standard replicator equations. N=1000, iterations=1,000,000

5 Interaction processes simulations

To analyze the effects of games in finite populations it is necessary to define microscopic update processes for how the agents in the populations interact and also compute the probabilities for an agent of one strategy to switch to another when they come into contact in the simulation.

In order to achieve this I focus on 3 different interaction processes previously studied, inspired from biology namely the Moran process, the Local update process and the Fermi process.

5.1 Transition probabilities

For the dynamics in a finite population, let i, j, k, and $N - i - j - k$, be the be the discrete counts of members of each strategy respectively.

5.1.1 Moran

In the Moran process at each time step an individual is chosen for reproduction proportional to their fitness within the population. Interestingly this process requires that each agent has access to some global average population fitness $\langle \pi \rangle$ since they are selected proportional to the average population fitness and not individual interactions with a single agent. An agent is also chosen uniformly at random to be killed and replaced by the agent selected for reproduction.

It has been shown that in the battle of sexes / matching pennies game, that this choice of process leads to a drift reversal effect in finite populations where the local update process does not due to extra term of N in the stochastic dynamics from a fokker planck equation. [4]

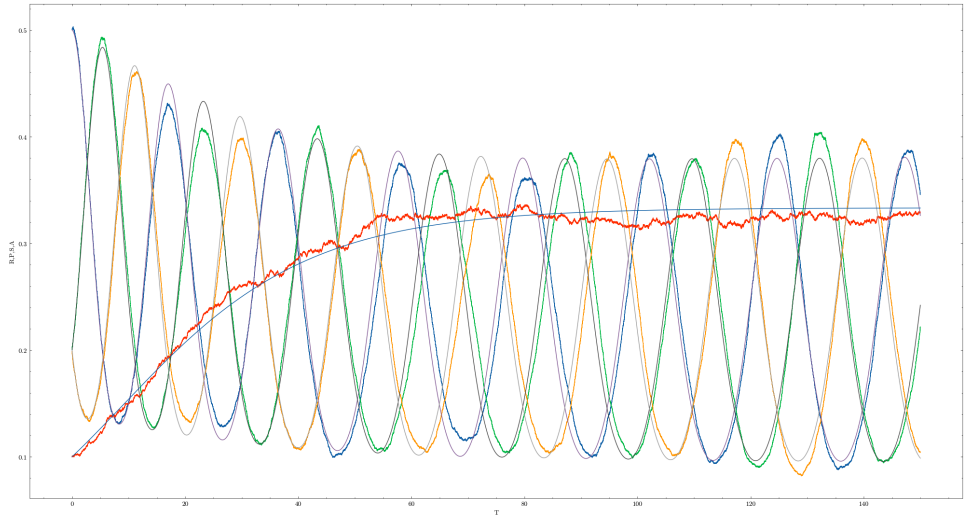


Figure 2: population size 60000 Moran process, along with numerical solution to adjusted dynamics.

For the Moran process, for a single transition $(R \to +)$ the transition probabilities are as follows:

$$T^{R\to+} = \frac{1}{2}\frac{1-w+w\pi_+}{1-w+w\langle\pi\rangle}\frac{i}{N}\frac{N-i-j-k}{N} \tag{21}$$

$$T^{+\to R} = \frac{1}{2}\frac{1-w+w\pi_R}{1-w+w\langle\pi\rangle}\frac{i}{N}\frac{N-i-j-k}{N} \tag{22}$$

Also transitions within RPS

$$T^{R\to S} = \frac{1}{2}\frac{1-w+w\pi_S}{1-w+w\langle\pi\rangle}\frac{i}{N}\frac{k}{N} \tag{23}$$

$$T^{S\to R} = \frac{1}{2}\frac{1-w+w\pi_R}{1-w+w\langle\pi\rangle}\frac{i}{N}\frac{k}{N} \tag{24}$$

The rest of the transitions, $T^{R\to P}, T^{P\to R}, T^{S\to P}, T^{P\to S}, T^{P\to +}, T^{+\to P}, T^{S\to +}, T^{+\to S}$ formed from permuations $(R, P, S, +)$ and $(i, j, k, N-i-j-k)$

The parameter w represents the selection pressure which determines how much the transition probabilities depend on the fitness differences between individuals. Weak selection increases the impact of stochastic drift in the dynamics and is often the biologically relevant case.

5.1.2 Local update

For the Local update process at each time step 2 randomly selected individuals a, b are chosen for reproduction. Individual b takes individual a's strategy proportional to the difference between their individual payoffs. Note that there is not global average payoff here, so this could be a more realistic assumption in nature where individuals would not have this knowledge.

Individual b switches to individual b's strategy with the probability:

$$p = \frac{1}{2} + \frac{w}{2}\frac{\pi_a - \pi_b}{\Delta\pi_{max}} \tag{25}$$

Where $\Delta\pi_{max}$ is the maximum possible difference in payoff between two individuals.

It follows that the transition probabilities for the populations in the combined game can be defined as:

$$T^{R\to+} = \left(\frac{1}{2} + \frac{w}{2}\frac{\pi_+ - \pi_R}{\Delta\pi_{\max}}\right)\frac{i}{N}\frac{N-i-j-k}{N} \tag{26}$$

for the case of $R \to +$, and the rest can again be written as permutations of $R, P, S, +$, and $i, j, k, N-i-j-k$.

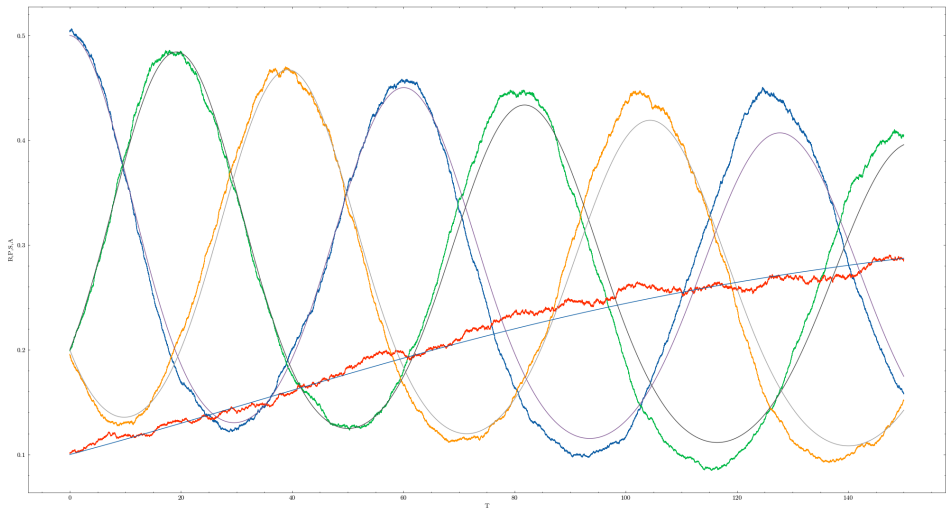


Figure 3: Time series of local update with population 20000, simulation trajectory normalized, compared with the numerical solution of the replicator equations derived from FP equation [4]

5.1.3 Fermi process

Similarly to the local update process two individuals are selected at each time step and individual b taks individual a's strategy with the probability:

$$p = \frac{1}{1 + e^{-w(\pi_a - \pi_b)}} \tag{27}$$

The transition probabilities:

$$T^{R\to+} = \left(\frac{1}{1 + e^{-w(\pi_a - \pi_b)}}\right)\frac{i}{N}\frac{N-i-j-k}{N} \tag{28}$$

5.1.4 General form

Generally the transition probabilites for any of the interaction processes $(R \to +)$ can be written as:

$$\phi(b \to a)\frac{N_a N_b}{N^2} \tag{29}$$

Given an appropriate ϕ reproductive function which are defined for each process.

$$\phi_{MO}(\pi_a, \langle\pi\rangle, w) = \frac{1}{2}\frac{1 - w + w\pi_a}{1 - w + w\langle\pi\rangle} \tag{30}$$

$$\phi_{LU}(\pi_a, \pi_b, w) = \frac{1}{2} + \frac{w}{2}\frac{\pi_a - \pi_b}{\Delta\pi_{max}} \tag{31}$$

$$\phi_{FP}(\pi_a, \pi_b, w) = \frac{1}{1 + e^{-w(\pi_a - \pi_b)}} \tag{32}$$

```
# Returns the transition probabilities given a repoductive function and payoffs.
# Returns dict indexed by T_RS for example between Rock and Scissors
# The 4th strategy is labelled as L
def transition_probs(reproductive_func, payoffs : list):
  # x y z, q = 1 - x - y - z

  names = ["R", "P", "S", "L"]
  freqs = [x, y, z, q]

  return {
      f"T_{a}{b}": reproductive_func(payoffs, i, j, w_sym) * freqs[i] * freqs[j]
      for i, a in enumerate(names)
      for j, b in enumerate(names)
      if i != j
  }
```

I have implemented this method in *aug_rps.py*, which can compute these transition probabilites symbolically for both derivation of larger equations and also for numerical method solutions later on. It takes a reproduction function ϕ for a list of payoffs for each strategy. The repoductive functions have to be defined in a general way, so take the payoffs of all strategies, then the index i, j which determines the strategies of individuals a, b described above and w.

```
def local_reproductive_func(payoffs, i, j, w):
  diff = payoffs[j] - payoffs[i]
  return (1/2) + (w / 2) * (diff / delta_pi)

def moran_reproductive_func(payoffs, i, j, w):
  payoff = payoffs[j]
  numerator = 1 - w + w * payoff
  denominator = 1 - w + w * averagePayoff
  return numerator / denominator

def fermi_reproductive_func(payoffs, i, j, w):
  diff = payoffs[j] - payoffs[i]
  return 1 / (1 + sp.exp((-w) * (diff)))
```

These are easily defined from the formal equations, so more can be added with new simple methods. Average payoff accessible globally for use in other formulas. The methods in this file are used for calculating exact values for the derived formula to complement the simulated values and also to produce exact deterministic trajectories in the infinite population size case very efficiently as running the simulations for very large N is very computationally expensive. These transitions can be used to numerically solve the adjusted replicator dynamics for each of these processes and used to plot deterministic trajectories (see more when plotting trajectories below.)

5.1.5 Master equation

Derivation of the fokker planck equation for these dynamics describing the evolution of the probability density function and eventually leading us to the adjusted dynamics for the different processes. It will become clear that the local update process is equivalent to the traditionall payoff difference replicator dynamics derived at the top of this paper, and we will also get deterministic equations for the Moran and fermi process plus the noise terms which depend on the value of N ultimately leading us to the drift reversal cases. The value computed for $a(x)$ will be the determinitic replicator dynamics allowing for numerical solutions of deterministic trajectories which can then be plotted as comparisons for the simulations, can also see that as N in simulations increase, the trajectories match perfectly as $N \to \infty$.

Initially for the trajectories I directly computed the adjusted factor for each process as in [4], however after deriving the FP equation it is possible to just store a global value for $a(x)$ which includes the scaling, and then the dynamics

can be computed and returned based purely on the transition probabilities which can be calculated numerically from the functions in the above section.
Master equation for Rock

$$\begin{aligned} P^{\tau+1}(i) - P^{\tau}(i) = {} & P^{\tau}(i-1)T^{PR}(i-1) - P^{\tau}(i)T^{PR}(i) + P^{\tau}(i-1)T^{SR}(i-1) - P^{\tau}(i)T^{SR}(i) \\ & + P^{\tau}(i-1)T^{+R}(i-1) - P^{\tau}(i)T^{+R}(i) + P^{\tau}(i+1)T^{RP}(i+1) - P^{\tau}(i)T^{RP}(i) \\ & + P^{\tau}(i+1)T^{RS} - P^{\tau}(i)T^{RS}(i) + P^{\tau}(i+1)T^{R+} - P^{\tau}(i)T^{R+}(i) \end{aligned} \tag{33}$$

Where i is the number of rock players.

$$\begin{aligned} T(x \pm \frac{1}{N}) &= T(x) \pm \frac{1}{N}T'(x) + \frac{1}{2N^2}T''(x)\dots \\ P(x \pm \frac{1}{N}) &= P(x) \pm \frac{1}{N}P'(x) + \frac{1}{2N^2}P''(x)\dots \end{aligned}$$

$$\begin{aligned} P^{\tau+1}(i) - P^{\tau}(i) = {} & (P - \frac{1}{N}P' + \frac{1}{2N^2}P'')(T^{PR} - \frac{1}{N}T^{PR'} + \frac{1}{2N^2}T^{PR''}) - PT^{PR} \\ & + (P - \frac{1}{N}P' + \frac{1}{2N^2}P'')(T^{SR} - \frac{1}{N}T^{SR'} + \frac{1}{2N^2}T^{SR''}) - PT^{SR} \\ & + (P - \frac{1}{N}P' + \frac{1}{2N^2}P'')(T^{+R} - \frac{1}{N}T^{+R'} + \frac{1}{2N^2}T^{+R''}) - PT^{+R} \\ & + (P + \frac{1}{N}P' + \frac{1}{2N^2}P'')(T^{RP} + \frac{1}{N}T^{RP'} + \frac{1}{2N^2}T^{RP''}) - PT^{RP} \\ & + (P + \frac{1}{N}P' + \frac{1}{2N^2}P'')(T^{RS} + \frac{1}{N}T^{RS'} + \frac{1}{2N^2}T^{RS''}) - PT^{RS} \\ & + (P + \frac{1}{N}P' + \frac{1}{2N^2}P'')(T^{R+} + \frac{1}{N}T^{R+'} + \frac{1}{2N^2}T^{R+''}) - PT^{R+} \end{aligned} \tag{34}$$

$$\begin{aligned} P^{\tau+1}(i) - P^{\tau}(i) = {} & -\frac{1}{N}\Big[P\big(T^{PR'} + T^{SR'} + T^{+R'} - T^{RP'} - T^{RS'} - T^{R+'}\big) \\ & \quad + P'\big(T^{PR} + T^{SR} + T^{+R} - T^{RP} - T^{RS} - T^{R+}\big)\Big] \\ & + \frac{1}{2N^2}\Big[P\big(T^{PR''} + T^{SR''} + T^{+R''} + T^{RP''} + T^{RS''} + T^{R+''}\big) \\ & \quad + 2P'\big(T^{PR'} + T^{SR'} + T^{+R'} + T^{RP'} + T^{RS'} + T^{R+'}\big) \\ & \quad + P''\big(T^{PR} + T^{SR} + T^{+R} + T^{RP} + T^{RS} + T^{R+}\big)\Big] \\ & = \frac{1}{N}\Big[-\big(P\big(T^{PR} + T^{SR} + T^{+R} - T^{RP} - T^{RS} - T^{R+}\big)\big)' \\ & + \frac{1}{2}\big(P\big(\frac{T^{PR} + T^{SR} + T^{+R} + T^{RP} + T^{RS} + T^{R+}}{N}\big)\big)''\Big] \end{aligned} \tag{35}$$

We then have for rock (i) $a(i) = T^{PR} + T^{SR} + T^{+R} - T^{RP} - T^{RS} - T^{R+}$ and $b^2(i) = \frac{T^{PR}+T^{SR}+T^{+R}+T^{RP}+T^{RS}+T^{R+}}{N}$.

For paper and scissors we get the same but with the corresponding ingoing and outgoing transition probabilities.

$$\begin{aligned} P^{\tau+1}(j) - P^{\tau}(j) = {} & \frac{1}{N}\Big[-\big(P\big(T^{RP} + T^{SP} + T^{+P} - T^{PR} - T^{PS} - T^{P+}\big)\big)' \\ & + \frac{1}{2}\big(P\big(\frac{T^{RP} + T^{SP} + T^{+P} + T^{PR} + T^{PS} + T^{P+}}{N}\big)\big)''\Big] \end{aligned} \tag{36}$$

$a(j) = T^{RP} + T^{SP} + T^{+P} - T^{PR} - T^{PS} - T^{P+}$ and $b^2(j) = \frac{T^{RP}+T^{SP}+T^{+P}+T^{PR}+T^{PS}+T^{P+}}{N}$.

$$P^{\tau+1}(k) - P^{\tau}(k) = \frac{1}{N}\Big[- \big(P\big(T^{RS} + T^{PS} + T^{+S} - T^{SR} - T^{SP} - T^{S+}\big)\big)' + \frac{1}{2}\big(P\big(\frac{T^{RS} + T^{PS} + T^{+S} + T^{SR} + T^{SP} + T^{S+}}{N}\big)\big)''\Big] \tag{37}$$

$a(k) = T^{RS} + T^{PS} + T^{+S} - T^{SR} - T^{SP} - T^{S+}$ and $b^2(k) = \frac{T^{RS}+T^{PS}+T^{+S}+T^{SR}+T^{SP}+T^{S+}}{N}$.

From these we can form the fokker planck equations for each strategies population ($x \in \{i, j, k\}$):

$$\frac{d}{dt}P^{\tau}(x) = \frac{1}{N}\Big[- \frac{d}{dx}\big(a(x)P^{\tau}(x)\big) + \frac{1}{2}\frac{d^2}{dx^2}\big(b^2(x)P^{\tau}(x)\big)\Big] \tag{38}$$

Where $a(x)$ corresponds to the deterministic replicator dynamics as previously defined e.g. the payoff comparison dynamics, and $b(x)$ is a noise correction term influenced by the transition probabilities and the population size N.

This results in the following Langevin equation with noise term $b(x)\xi$ (stochastic differential equation):

$$\dot{x} = a(x) + b(x)\xi \tag{39}$$

$a(x)$ is the deterministic replicator equation as $N \to \infty$ noise dissapears ($b^2(x)$ is divided by N).

5.2 Link to deterministic equations $\lim_{N\to\infty}$

Adjusted deterministic equations for the interaction processes, derived from the fokker planck expansions of the interaction processes defined (master equation).

In the standard case where N is very large or $N \to \infty$, the noise term dissapears as it has a factor of $\sqrt{\frac{1}{N}}$ which leaves just:

$$\dot{x} = a(x) \tag{40}$$

For the Moran process, the infinite population dynamics reduce to a scaled version of the ordinary replicator equation by the factor $\frac{1}{\Gamma+\langle\pi(x)\rangle}$

$$a(x) = \frac{1}{\Gamma + \langle\pi(x)\rangle} x_i(\pi_i(x) - \langle\pi(x)\rangle), \Gamma = \frac{1-w}{w} \tag{41}$$

$$\begin{aligned}\langle\pi(x)\rangle = {} & x\left(ax + bz + cy + \gamma\left(-x - y - z + 1\right)\right) \\ & + y\left(ay + bx + cz + \gamma\left(-x - y - z + 1\right)\right) \\ & + z\left(az + by + cx + \gamma\left(-x - y - z + 1\right)\right) \\ & + \left(x\left(a + \beta\right) + y\left(a + \beta\right) + z\left(a + \beta\right)\right)\left(-x - y - z + 1\right)\end{aligned}$$

5.3 Constant of motion ΔH definition (drift)

5.3.1 $\langle \Delta H_{SD} \rangle$ Derivation

The equation $H = -q(1-q)$ defines a constant of motion for the SD part of the game, where q is the fraction of players in the 4th strategy. Using the transition probabilities of the different process we can derive an expression for the expected change in H within the simplex.

Where $i, j, k, N-i-j-k$ are the players playing R, P, S, and the 4th strategy respectively.

$\Delta H = H(t+1) - H(t)$,
$\Delta H = -x_{t+1}(1-x_{t+1}) - (-x_t(1-x_t))$
$\Delta H = -x_{t+1}(1-x_{t+1}) + x_t(1-x_t)$
$\Delta H = x_t(1-x_t) - x_{t+1}(1-x_{t+1})$

$\langle \Delta H \rangle = \sum_{i,j,k} \Big(H_s - H_{s'}\Big) T^{s \to s'}$, s is a particular state in the simplex.

$$\begin{aligned}
\langle \Delta H_{SD} \rangle = \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& (N-i-j-k)(1-N+i+j+k)(T^{R+} + T^{P+} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) \\
& - (N-i-j-k+1)(-N+i+j+k)T^{R+} \\
& - (N-i-j-k+1)(-N+i+j+k)T^{P+} \\
& - (N-i-j-k-1)(2-N+i+j+k)T^{+R} \\
& - (N-i-j-k-1)(2-N+i+j+k)T^{+P} \\
& - (N-i-j-k-1)(2-N+i+j+k)T^{+S} \Big]
\end{aligned} \tag{42}$$

The terms with transitions within the RPS simplex can be ignored as q would not change between these states, therefore the term $H_s - H_{s'} = 0$

$p = N-i-j-k$, the number of players playing the 4th strategy.

$$\begin{aligned}
\langle \Delta H_{SD} \rangle = \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& p(1-p)(T^{R+} + T^{P+} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) \\
& - (p+1)(-p)(T^{R+} + T^{P+} + T^{S+}) \\
& - (p-1)(2-p)(T^{+R} + T^{+P} + T^{+S}) \Big]
\end{aligned}$$

$$\begin{aligned}
\langle \Delta H_{SD} \rangle &= \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[T^{R+} \left(p(1-p) + p(p+1)\right) + T^{+R} \left(p(1-p) - (p-1)(2-p)\right) \\
&\quad + T^{P+} \left(p(1-p) + p(p+1)\right) + T^{+P} \left(p(1-p) - (p-1)(2-p)\right) \\
&\quad + T^{S+} \left(p(1-p) + p(p+1)\right) + T^{+S} \left(p(1-p) - (p-1)(2-p)\right) \Big] \\
&= \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[2p \left(T^{R+} + T^{P+} + T^{S+}\right) + (p - p^2 - (3p - p^2 - 2))(T^{+R} + T^{+P} + T^{+S}) \Big] \\
&= \frac{6}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[2p \left(T^{R+} + T^{P+} + T^{S+}\right) + (2-2p)(T^{+R} + T^{+P} + T^{+S}) \Big]
\end{aligned}$$

$$\langle \Delta H_{SD} \rangle = \frac{12}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[p \left(T^{R+} + T^{P+} + T^{S+} \right) + (1-p)(T^{+R} + T^{+P} + T^{+S}) \Big]$$

$$\langle \Delta H_{SD} \rangle = \frac{12}{N^5} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} p \Big[(T^{R+} - T^{+R}) + (T^{P+} - T^{+P}) + (T^{S+} - T^{+S}) \Big] + T^{+R} + T^{+P} + T^{+S} \tag{43}$$

The continuous limit, where $x = i/N$, $y = j/N$, $z = k/N$, and $q = p/N$, $p = Nq$ and $q = 1 - x - y - z$ leads to:

$$\begin{aligned} \langle \Delta H_{SD} \rangle = \frac{12}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& Nq \left[(T^{R+} - T^{+R}) + (T^{P+} - T^{+P}) + (T^{S+} - T^{+S}) \right] \\ & + (T^{+R} + T^{+P} + T^{+S}) \Big] \end{aligned}$$

Finally,

$$\begin{aligned} \langle \Delta H_{SD} \rangle = & \frac{12}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[q \left[(T^{R+} - T^{+R}) + (T^{P+} - T^{+P}) + (T^{S+} - T^{+S}) \right] \Big] \\ & + \frac{12}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \left[T^{+R} + T^{+P} + T^{+S} \right] \end{aligned} \tag{44}$$

This can then be solved numerically and the critical population values can be found where $\langle \Delta H_{SD} \rangle = 0$.

Critical N found matches nicely with the simulated versions. Numerical integration in python code ./augRps.py, shows change of sign as expected. Matches nicely with the approximated values for the Moran process. The specific expression for Moran process is very long. Computed numerically and solved with scipy.integrate (reference scipy) Maybe can plot the simulated critical population sizes against the analyical on the same graph.

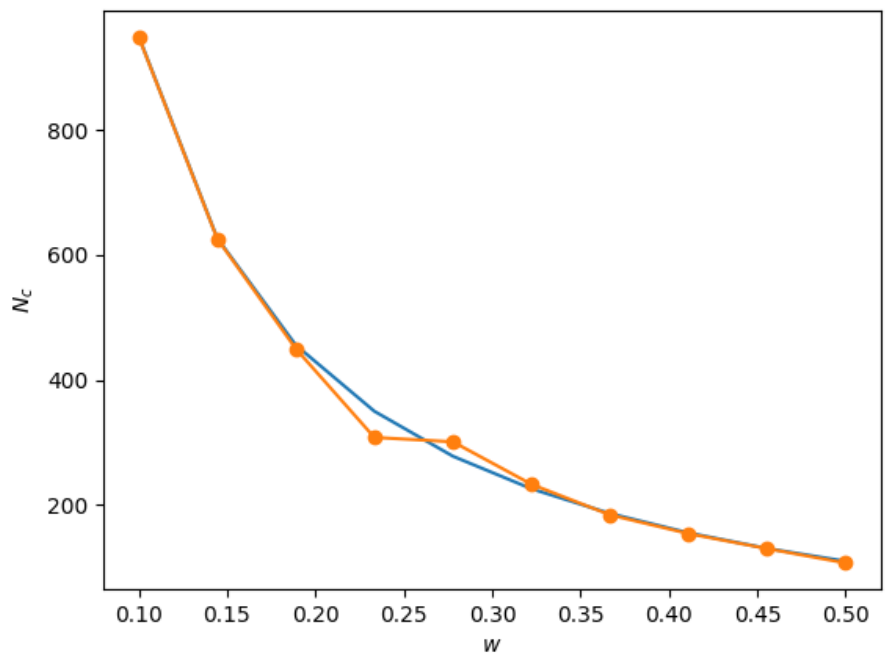


Figure 4: Comparison of simulated critical population sizes against the derived expected change in H for Moran process. Small deviations due to stochasticity in the simulations. $\gamma = 0.2, \beta = 0.1$ standard RPS.

5.3.2 $\langle \Delta H_{RPS} \rangle$ Derivation

$\langle \Delta H \rangle$ within the RPS plane $H = -xyz$.

$$\begin{aligned}\langle \Delta H_{RPS} \rangle = \frac{2}{N^6} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& ijk(T^{RP} + T^{RS} + T^{R+} + T^{PR} + T^{PS} + T^{P+} \\ & + T^{SR} + T^{SP} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) \\ & - k(i-1)(j+1)T^{RP} - (i-1)j(k+1)T^{RS} - jk(i-1)T^{R+} \\ & - k(i+1)(j-1)T^{PR} - i(j-1)(k+1)T^{PS} - ik(j-1)T^{P+} \\ & - (i+1)j(k-1)T^{SR} - i(j+1)(k-1)T^{SP} - ij(k-1)T^{S+} \\ & - jk(i+1)T^{+R} - ik(j+1)T^{+P} - ij(k+1)T^{+S} \Big]\end{aligned}$$

$$\begin{aligned}\langle \Delta H_{RPS} \rangle = \frac{2}{N^6} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& k(j-i)(T^{RP} - T^{PR}) + j(k-i)(T^{RS} - T^{SR}) + i(j-k)(T^{SP} - T^{PS}) \\ & + k(T^{RP} + T^{PR}) + j(T^{RS} + T^{SR}) + i(T^{SP} + T^{PS}) \\ & + jk(T^{R+} - T^{+R}) + ik(T^{P+} - T^{+P}) + ij(T^{S+} - T^{+S}) \Big]\end{aligned} \tag{45}$$

Compared to the derivation of simply the RPS case in[1], this has the additional terms including the difference in tranition probabilities in and out of the 4th strategy.

$$
\begin{aligned}
\langle \Delta H_{RPS} \rangle = \frac{2}{N^3} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& N^2 z(y-x)(T^{RP} - T^{PR}) + N^2 y(z-x)(T^{RS} - T^{SR}) \\
& + N^2 x(y-z)(T^{SP} - T^{PS}) \\
& + Nz(T^{RP} + T^{PR}) + Ny(T^{RS} + T^{SR}) + Nx(T^{SP} + T^{PS}) \\
& + N^2 xz(T^{P+} - T^{+P}) + N^2 yz(T^{R+} - T^{+R}) + N^2 xy(T^{S+} - T^{+S}) \Big]
\end{aligned}
$$

$$
\begin{aligned}
\langle \Delta H_{RPS} \rangle = \frac{2}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& z(y-x)(T^{RP} - T^{PR}) + y(z-x)(T^{RS} - T^{SR}) \\
& + x(y-z)(T^{SP} - T^{PS}) + xz(T^{P+} - T^{+P}) + yz(T^{R+} - T^{+R}) + xy(T^{S+} - T^{+S}) \Big] \\
& + \frac{2}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[z(T^{RP} + T^{PR}) + y(T^{RS} + T^{SR}) + x(T^{SP} + T^{PS}) \Big]
\end{aligned} \tag{46}
$$

Rough figures of rps and SD delta H values.

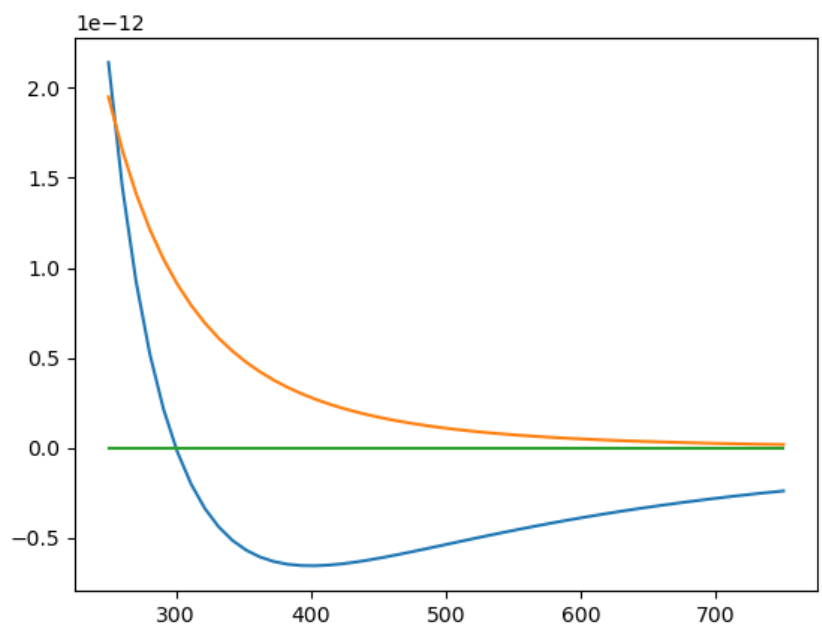


Figure 5: Blue - SD

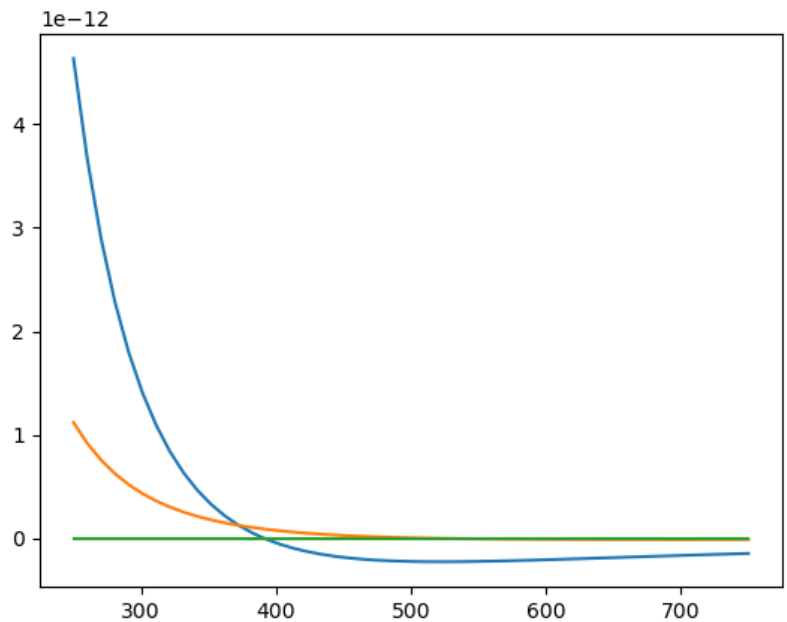


Figure 6: Double reversal case. Blue - SD

5.3.3 $\langle \Delta H_4 \rangle$ Derivation

All 4 strategies, $H = -xyz(1-x-y-z)$, Therefore $\langle \Delta H_4 \rangle$ for the population counts can be defined as:

$$\begin{aligned}\langle \Delta H_4 \rangle = \frac{6}{N^7} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& ijk(N-i-j-k)(T^{RP} + T^{RS} + T^{R+} + T^{PR} + T^{PS} + T^{P+} \\ & + T^{SR} + T^{SP} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) - (i-1)(j+1)k(N-i-j-k)T^{RP} \\ & - (i-1)j(k+1)(N-i-j-k)T^{RS} - (i-1)jk(N-i-j-k+1)T^{R+} \\ & - (i+1)(j-1)k(N-i-j-k)T^{PR} - i(j-1)(k+1)(N-i-j-k)T^{PS} \\ & - i(j-1)k(N-i-j-k+1)T^{P+} - (i+1)j(k-1)(N-i-j-k)T^{SR} \\ & - i(j+1)(k-1)(N-i-j-k)T^{SP} - ij(k-1)(N-i-j-k+1)T^{S+} \\ & - (i+1)jk(N-1-i-j-k)T^{+R} - i(j+1)k(N-1-i-j-k)T^{+P} - ij(k+1)(N-1-i-j-k)T^{+S} \Big]\end{aligned} \tag{47}$$

Normalization $\frac{6}{N^7}$, as its over the whole simplex (pyramid volume 1/6) and triple summation, (N^3), then 4 populations N^4.

Let p = N - i - j - k (4th strategy count):

$$
\begin{aligned}
\langle \Delta H_4 \rangle = \frac{6}{N^7} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& ijkp(T^{RP} + T^{RS} + T^{R+} + T^{PR} + T^{PS} + T^{P+} \\
& + T^{SR} + T^{SP} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) - (i-1)(j+1)kpT^{RP} \\
& - (i-1)j(k+1)pT^{RS} - (i-1)jk(p+1)T^{R+} \\
& - (i+1)(j-1)kpT^{PR} - i(j-1)(k+1)pT^{PS} \\
& - i(j-1)k(p+1)T^{P+} - (i+1)j(k-1)pT^{SR} \\
& - i(j+1)(k-1)pT^{SP} - ij(k-1)(p+1)T^{S+} \\
& - (i+1)jk(p-1)T^{+R} - i(j+1)k(p-1)T^{+P} - ij(k+1)(p-1)T^{+S} \Big]
\end{aligned} \tag{48}
$$

Looking at one pair of transitions in and out of the same state. $(R \to P, P \to R)$ this can be rewritten as before for $\langle \Delta H_{RPS} \rangle$ as differences of transition probabilities in and out of a state.

$$
\begin{aligned}
\langle \Delta H_4 \rangle = \frac{6}{N^7} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& kp(j-i)(T^{RP} - T^{PR}) + jp(k-i)(T^{RS} - T^{SR}) \\
& + ip(j-k)(T^{SP} - T^{PS}) + kp(T^{RP} + T^{PR}) + jp(T^{RS} + T^{SR}) \\
& + ip(T^{SP} + T^{PS}) + jk(p-i)(T^{R+} - T^{+R}) + ik(p-j)(T^{P+} - T^{+P}) \\
& + ij(p-k)(T^{S+} - T^{+S}) + jk(T^{R+} + T^{+R}) + ik(T^{P+} + T^{+P}) + ij(T^{S+} + T^{+S}) \Big]
\end{aligned} \tag{49}
$$

As before, converting this to the integral form where $x = i/N, y = j/N, z = k/N$ and $q = p/N, p = Nq$ where $q = 1 - x - y - z$ leads to:

$$
\begin{aligned}
\langle \Delta H_4 \rangle = \frac{6}{N^4} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[& N^3 zq(y-x)(T^{RP} - T^{PR}) + N^3 yq(z-x)(T^{RS} - T^{SR}) \\
& + N^3 xq(y-z)(T^{SP} - T^{PS}) + N^2 zq(T^{RP} + T^{PR}) + N^2 yq(T^{RS} + T^{SR}) + N^2 xq(T^{SP} + T^{PS}) \\
& + N^3 yz(q-x)(T^{R+} - T^{+R}) + N^3 xz(q-y)(T^{P+} - T^{+P}) + N^3 xy(q-z)(T^{S+} - T^{+S}) \\
& + N^2 yz(T^{R+} + T^{+R}) + N^2 xz(T^{P+} + T^{+P}) + N^2 xy(T^{S+} + T^{+S}) \Big]
\end{aligned} \tag{50}
$$

$$
\begin{aligned}
\langle \Delta H_4 \rangle = & \frac{6}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[zq(y-x)(T^{RP} - T^{PR}) + yq(z-x)(T^{RS} - T^{SR}) + xq(y-z)(T^{SP} - T^{PS}) \\
& + yz(q-x)(T^{R+} - T^{+R}) + xz(q-y)(T^{P+} - T^{+P}) + xy(q-z)(T^{S+} - T^{+S}) \Big] \\
& + \frac{6}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[zq(T^{RP} + T^{PR}) + yq(T^{RS} + T^{SR}) + xq(T^{SP} + T^{PS}) \\
& + yz(T^{R+} + T^{+R}) + xz(T^{P+} + T^{+P}) + xy(T^{S+} + T^{+S}) \Big]
\end{aligned} \tag{51}
$$

Final derivation of $\langle \Delta H_4 \rangle$. There are clear similarities to the other two expected values $\langle \Delta H_{SD} \rangle, \langle \Delta_{RPS} \rangle$ with some terms missing due to the different observation variable H used but they can all be rewritten in terms of differences of transition probabilities into and out of state (e.g. $(T^{RP} - T^{PR})$) in an integral scaled by $1/N$ plus positive terms in a second integral scaled by $1/N^2$.

5.3.4 Graphical comparison of H

Discuss how the H4 is a composition of the drift and diffusion terms of the other two expected delta H.

Point cloud - look at how H-RPS shows the attraction or repulsion of the fixed rps center, SD - attraction to the top point or the bottom RPS pure plane,

Then H4 - combination where it sign change would indicate whether or not the system is attracted to the central internal fixed point in the pyramid - where a sign change would suggest it pushed to one of the faces e.g. the rps plane where q = 0, step towards the boundary means larger H4 - so delta H addition of larger value so sign flip...

Change in sign in SD - up or down Change in sign in RPS - into out out of center or RPS plane Change in sign in H4 - away from internal fixed point in any direction

Compare the 3 together on same plot for some of the interesting cases.

$\langle \Delta H_4 \rangle$ seems to have a sign change when either the RPS or SD game does or both, double check the code copy of the formula, and see if theres a pattern as one does not seem obvious at the moment.

6 Drift

$$\begin{bmatrix} 0 & -s & 1 & 0.2 \\ 1 & 0 & -s & 0.2 \\ -s & 1 & 0 & 0.2 \\ 0.1 & 0.1 & 0.1 & 0 \end{bmatrix} \tag{52}$$

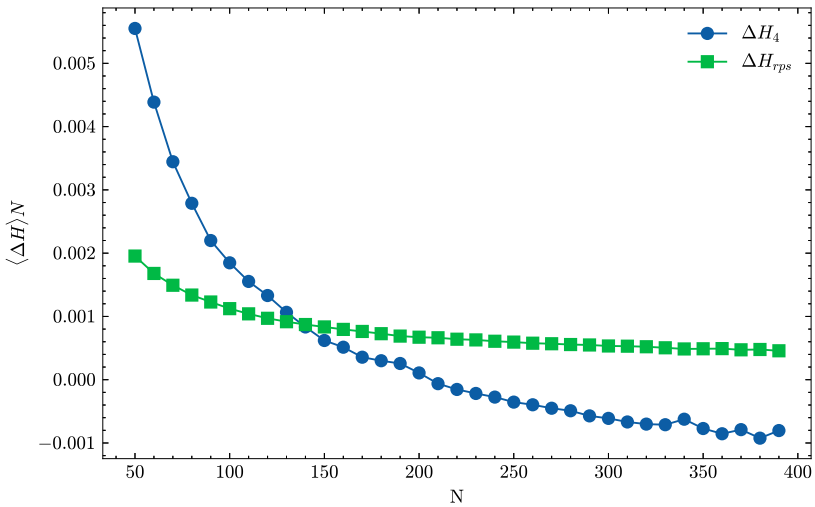


Figure 7: Drift reversal in the above game 52, in the 4th strategy $\langle \Delta H \rangle$ changes sign. Averages over 10,000,000 realisations, fixed $w = 0.45$, ΔH_{rps} approaches 0 but does not reverse, $s = 1$

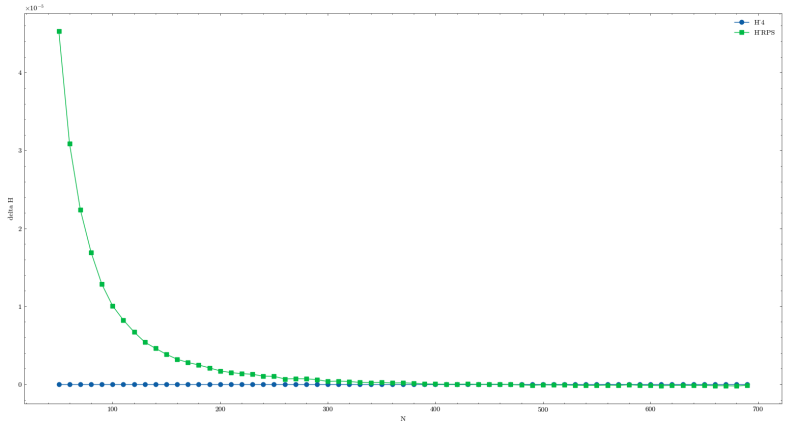


Figure 8: Drift reversal in the standard RPS game - 0 playing 4th strategy, $s = 0.8$, deterministic case the internal fixed point is stable ($\langle \Delta H_{rps} \rangle > 0$) but again it changes sign at a small population size, average over 10,000,000 realisations. Change of sign at $N \approx 470$

$$\begin{bmatrix} 0 & -s & 1 & 0.5 \\ 1 & 0 & -s & 0.5 \\ -s & 1 & 0 & 0.5 \\ 0+\beta & 0+\beta & 0+\beta & 0 \end{bmatrix} \tag{53}$$

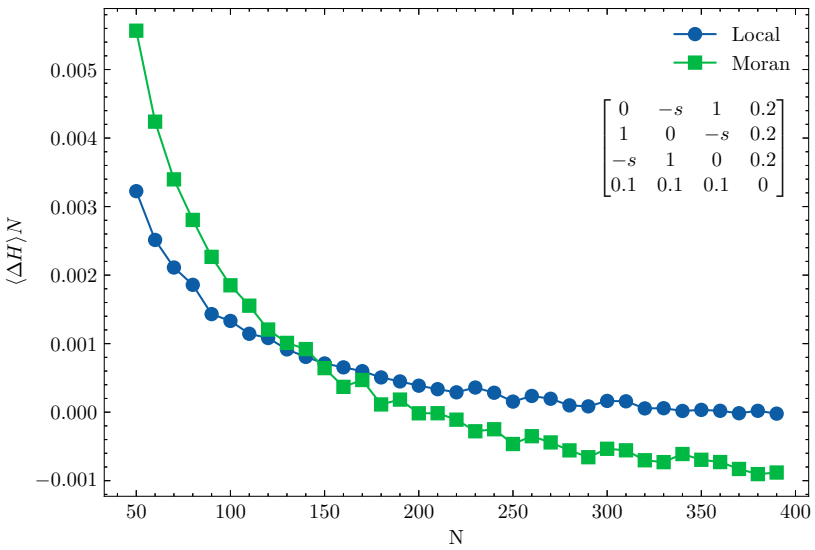


Figure 9: Moran and local update drift for ΔH_4 - 2×10^7 realizations , needs more average but drift reversal by N = 400, with w = 0.45

$$\begin{bmatrix} 0 & -s & 1 & \gamma \\ 1 & 0 & -s & \gamma \\ -s & 1 & 0 & \gamma \\ 0+\beta & 0+\beta & 0+\beta & 0 \end{bmatrix} \tag{54}$$

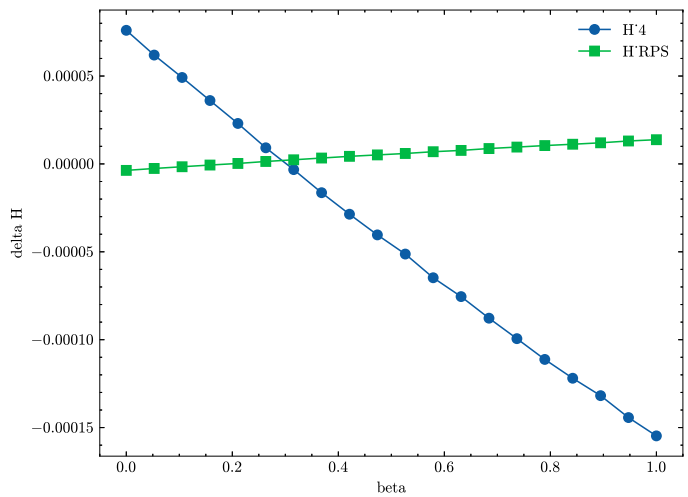


Figure 10: $\langle \Delta H_4 \rangle$ and $\langle \Delta H_{rps} \rangle$ for range of β values (0 to 1) in above matrix, fixed value of $\gamma = 0.5, w = 0.45, N = 200$

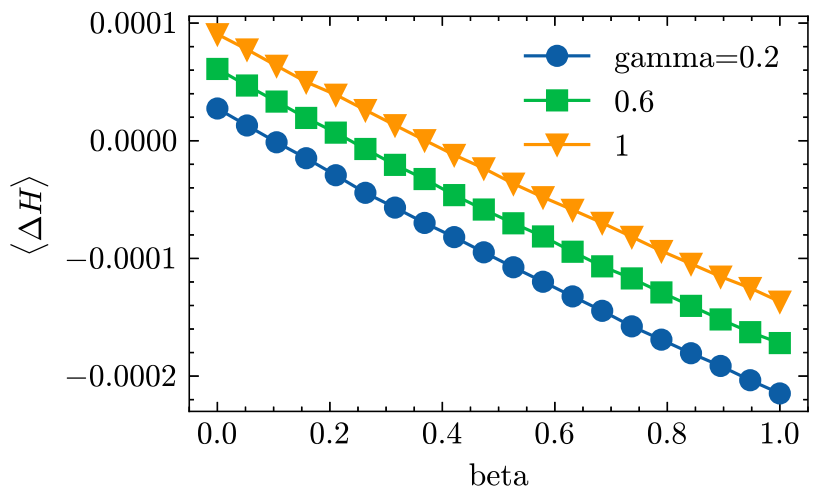


Figure 11: $\langle \Delta H_4 \rangle$ for different β, for 3 different γ

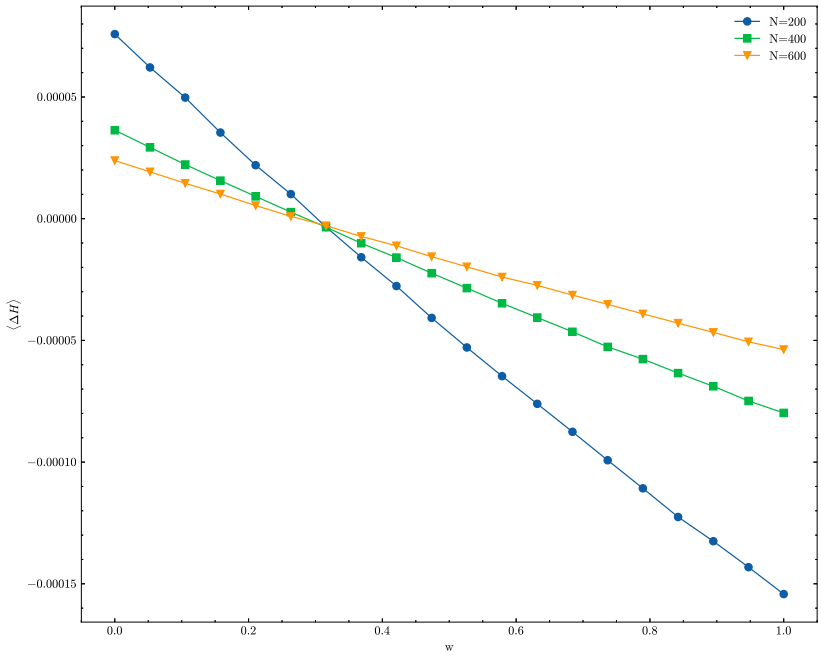


Figure 12: $\langle \Delta H_4 \rangle$ for different β at different pop sizes.

6.1 Critical population size N_c

The critical population size where drift occurs for a varying selection pressure w.
Implemented using the previous simulation code at suitably large simulation repetitions, binary search around the change of sign for the ΔH_4 value and repeated for range of w's.

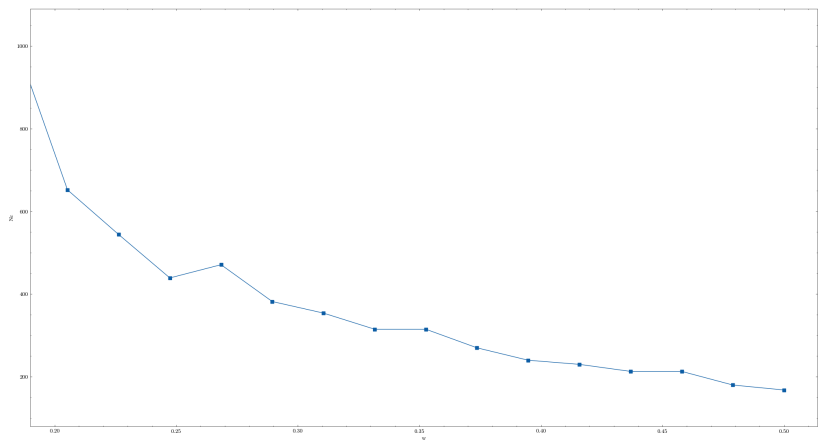


Figure 13: Critical N against W, need to average this with more realizations.

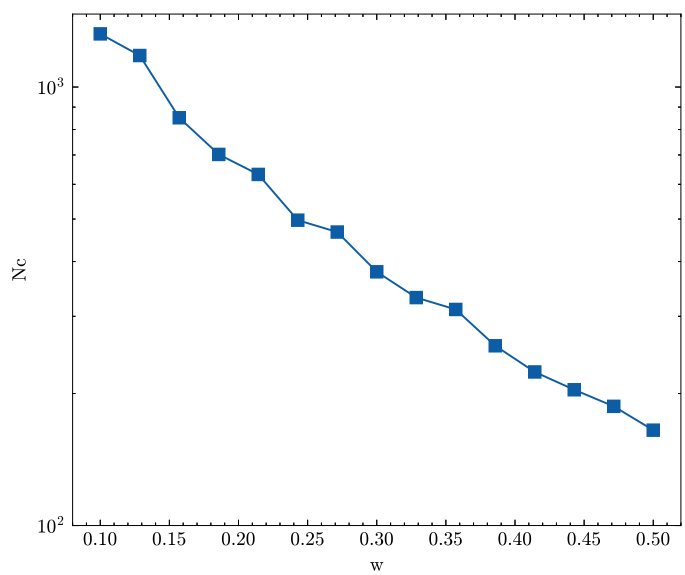


Figure 14: 5×10^7 realizations, critical N for drift reversal in 52

x axis label incorrect below

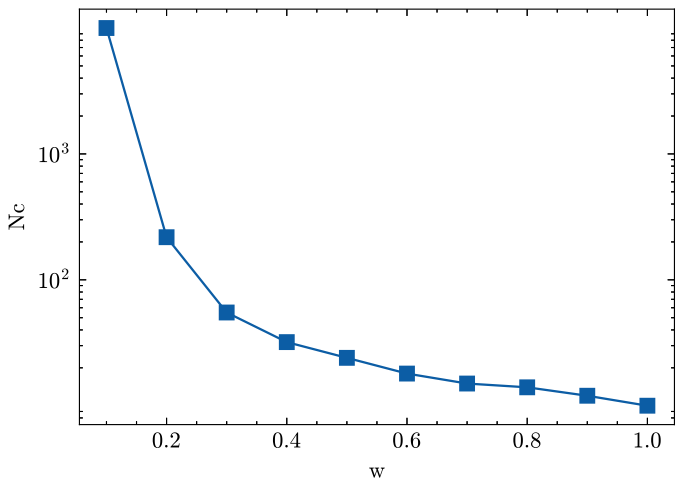


Figure 15: Critical population size N_c for range of β from 0 to 1, fixed $w = 0.45$, fixed $\gamma = 0.5$

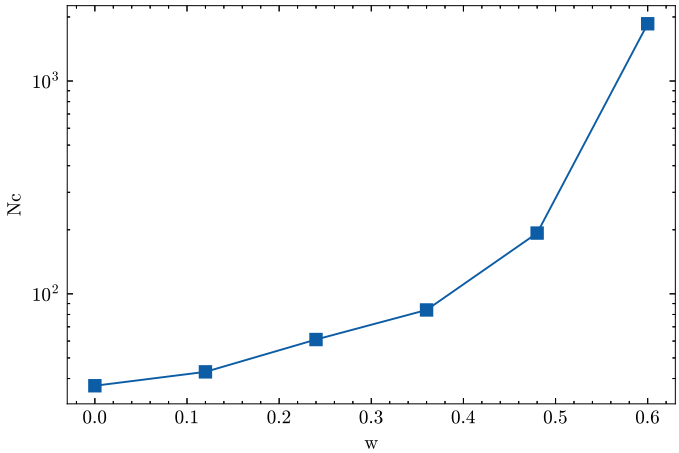
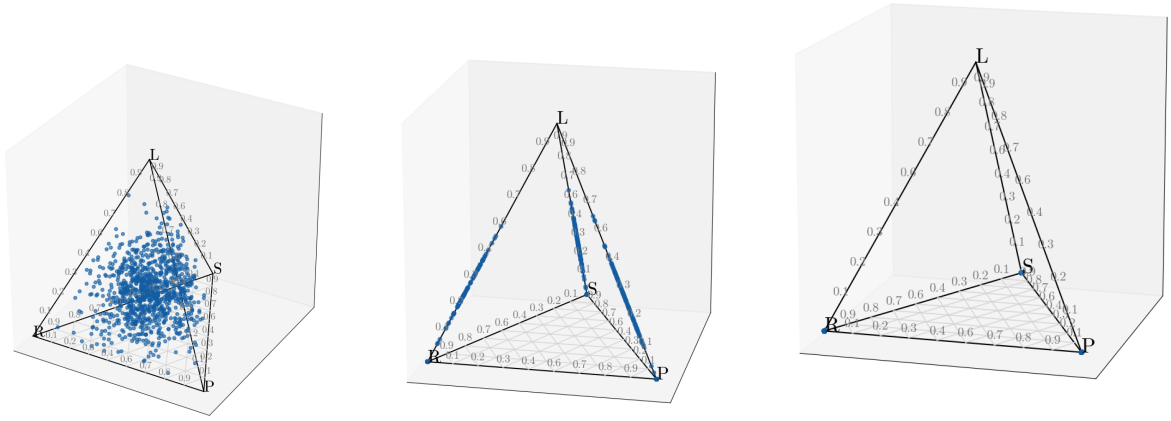


Figure 16: Critical population size N_c for range of γ from 0 to 1, fixed $w = 0.45$, fixed $\beta = 0.2$



(a) Initial

(b) N = 200, rps drift outwards

(c) N = 100, rps drift out and strategy 4 drift down

Figure 17: Initial random starting points, at low population we seee RPS drift occuring first, and the game is still at SD equilibrium, but at lower strategy e.g $N < 150$ as in the drift graph further up we seee reversal of both strategies to the bottom 3 corners, when $c > b$, or to all 4 corners when $b = c$ ($b = \beta, c = \gamma$ in the matrices in the document), similar results for both $s = 1, s = 0.8$ in the small population cases and both time RPS drifst out before SD and SD drifts at low population - only difference in large population case

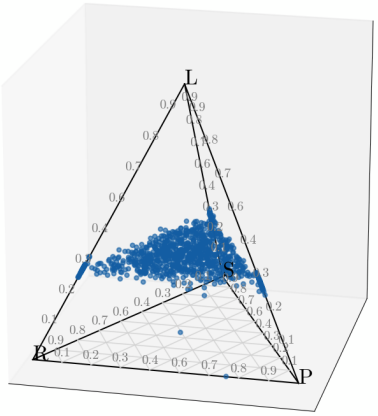


Figure 18: Large pop - fixes in orbit about center, s = 1 - very large population for neutrally stable RPS

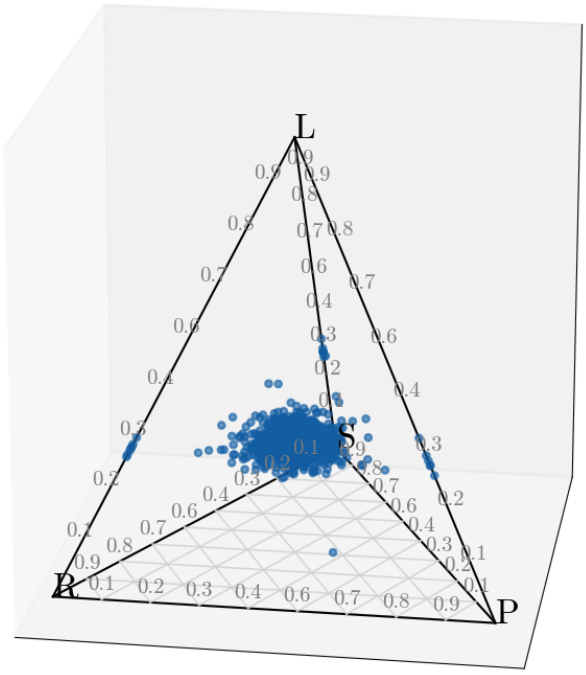


Figure 19: large pop, s = 0.8, converges to central fixed point. - large population for asymptotically stable RPS - $s < 1$

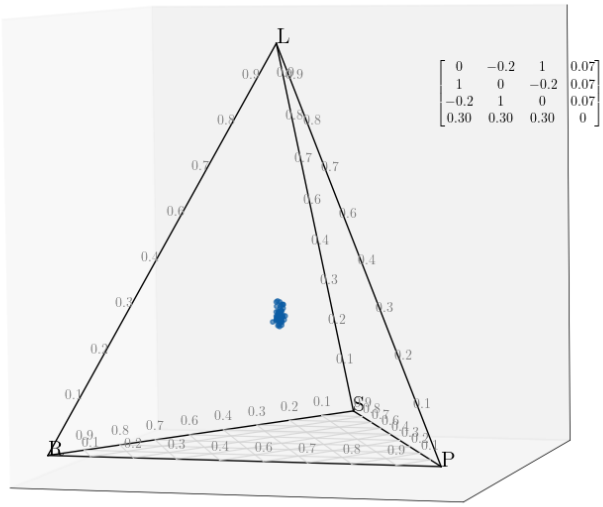
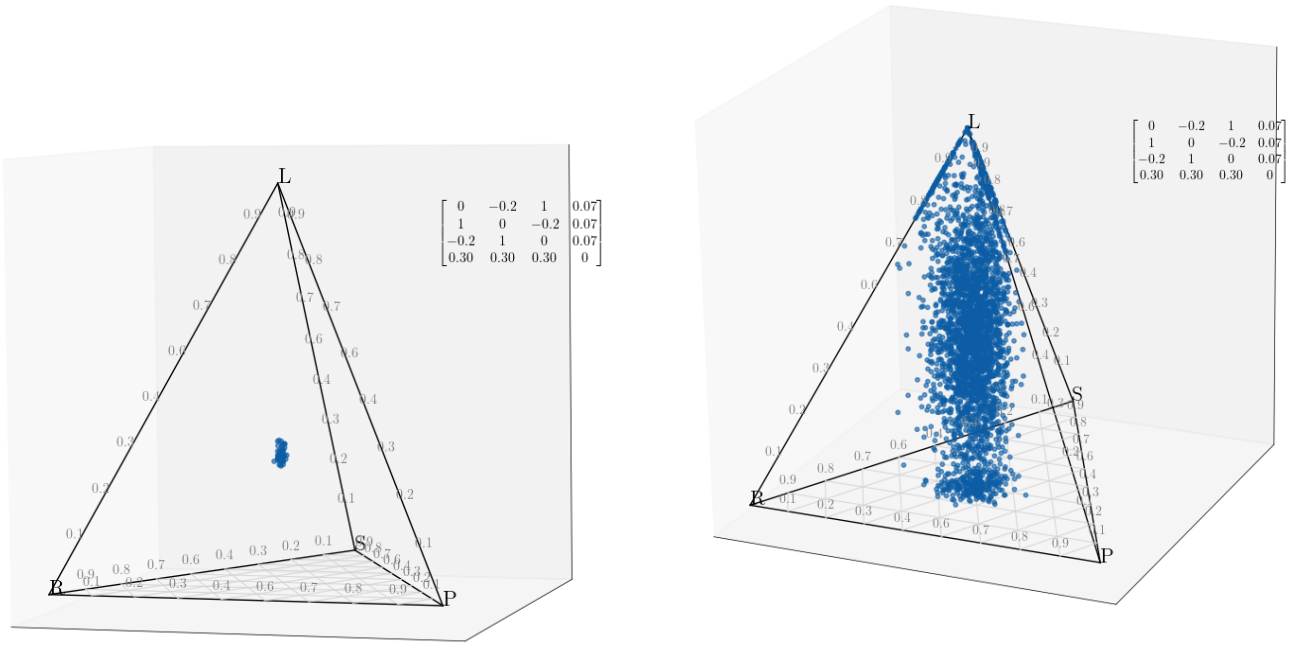


Figure 20: large pop, matrix in diagram



(a) $N = 50000$ (b) $N = 850$

Figure 21: $w = 0.45$, Strong attracting RPS with matrix in figure, reversal seen in SD but not in RPS, large population size tends to the deterministic case. - generate a $\Delta H_4, \Delta H_{RPS}$ plot for the above to show the drift reversal in SD game and not in RPS.

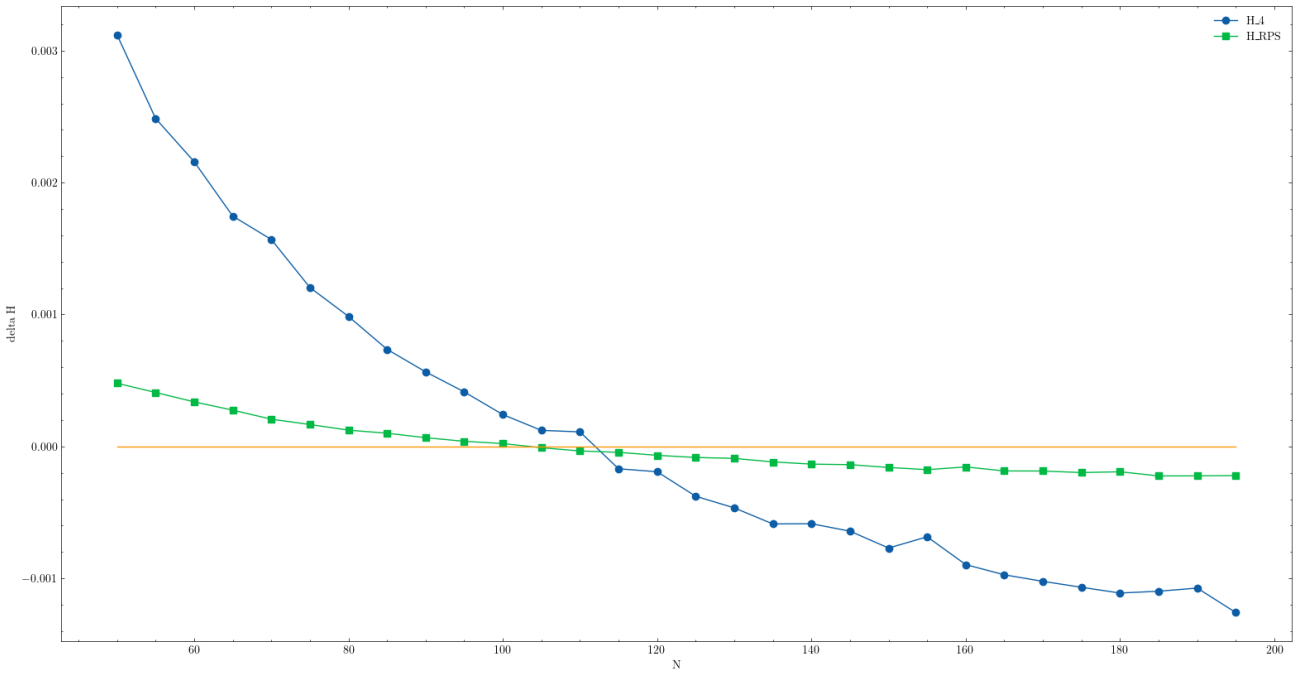


Figure 22: Same game used for above cloud - RPS crossing the x axis at a lower N_c - explains SD drift first

7 Refs

Scienceplots library for simple styling ref. [2]
Numba jit python compiler - used for large optimisations pre compiling intensive functions to fast machine code. [3]
Huge performance increase using numba in no-python mode compiling all the critical simulation methods into machine code and inlining them where appropriate to reduce function call overhead. Reached feasible simulation speed.
Drift reversal paper [4]

References

[1] Jens Christian Claussen and Arne Traulsen. Cyclic dominance and biodiversity in well-mixed populations. *Phys. Rev. Lett.*, 100:058104, Feb 2008.

[2] John D. Garrett. garrettj403/SciencePlots. September 2021.

[3] Siu Kwan Lam, Antoine Pitrou, and Stanley Seibert. Numba: A llvm-based python jit compiler. In *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, pages 1–6, 2015.

[4] Arne Traulsen, Jens Christian Claussen, and Christoph Hauert. Coevolutionary dynamics: From finite to infinite populations. *Phys. Rev. Lett.*, 95:238701, Dec 2005.